The Optimum Study — Comparison Table & Study Guide

For: MIDS graduate student prep. Everything here maps to concepts from W203 (statistics) and W241 (experiments & causal inference): potential outcomes, randomization, DiD, regression discontinuity, clustered errors, bootstrap. Learn the numbers cold, the mechanisms in your own words, and the scripts verbatim.

Part 1 · The Comparison Table

Where THEIR approach (the APR Calculator) is better

dimension	their approach	why it beats ours
Direct observation of the treatment	They measure `Q_mump2p(80%)` from their own gateways — actual mump2p delivery times	We cannot observe mump2p at all (shadow mode is invisible in public data). Our Δ is an assumed floor/multiplier; theirs is measured on the real product
Outcome completeness	Reduced-form: realized operator APR captures every revenue channel at once, including ones we didn't model	A structural model is only as complete as its author's imagination
Baseline anchor	2.91% network APR from Rated — realized yield	Realized APR is the better denominator for a "% uplift" claim than our spec-derived issuance
Reach threshold	One spec-motivated criterion: p80 propagation ≈ 66% attestation supermajority + margin	Cleaner than our patchwork (p90 spread, 40% proposer-boost, headroom caps)
Auditability	Δ × slope × stake — auditable in five minutes	Our pipeline takes days and a statistician. For a *calculator*, theirs is right
Updateability	Δ re-measures continuously from live telemetry	Our calibration is frozen to study windows

Where OUR study is better

dimension	their approach	why ours beats it
Causal identification	Cross-sectional APR-vs-latency across operators — confounded	Deadline dose-response + RANDAO random assignment: same validators, random exposure
Venue transfer	Hoodi Δ transplanted raw onto a mainnet slope	Floor model `min(observed, floor)`: 6.7× on Hoodi → 2.2× at mainnet's median
Equilibrium / adoption	"Mean-field assumptions"	Measured adoption curve: ~40% receiver ramp, (1−α) arms-race decay, sweet spot 40–55%, public-goods floor at saturation
Behavioral honesty	"Immediate revenue improvements"	92% of value requires re-tuned publication timing; passive install ≈ $2–3/yr
Physical constraints	Unbounded multiplier extrapolation	Light-speed floor; deadline headroom; 6×→20× moves the total only 11%
Falsification	None — no placebo, control, or null ever published	Five independent designs on the deployed network; placebos; 1× calibration checks; ~60 unit tests
Channel structure	One black-box slope	A/B/C/D separable with ownership; B/C exclusivity; orphan values validated vs relay payloads (1.02×)
Uncertainty	Point estimates	Scenarios everywhere: 6×/3×/2×, floors 150/250/400ms, cloud/metal
Reproducibility	Private operator panel, unseen figure, gateway telemetry	End-to-end public data, open pipeline
Framing integrity	Present-tense revenue for an undeployed product	Counterfactual labeling enforced in code, dashboard, report
Bandwidth channel	Absent	Modeled: ~3.8 TB/node/yr, priced per hosting regime

Synthesis line (say this): "*The approaches are complements. They hold the one thing I can't get — direct measurement of the product. I hold everything that makes a number believable — identification, transfer, equilibrium, falsification. And both methods converge on about thirty dollars per validator per year, so the fight isn't over the answer; it's over which method survives scrutiny getting there.*"

Part 2 · Numbers You Must Know Cold

number	what it is	one-line derivation
4,000 ms	attestation deadline (1/3 of a 12s slot)	consensus spec; attesters vote for whatever head they see at that instant
99.9% → 67.4% → 0.1%	correct-head rate: on-time / 4–5s / >6s	dose-response over 216,000 mainnet slots, attester-weighted
0.94%	measured head-miss rate	1 – weighted mean correct_head_rate
61%	share of head misses on ON-TIME blocks	tail nodes; the reason "rescue late blocks" is the wrong model
3,656 vs 1,811 ms	median *publication* time, late vs normal blocks	late blocks are published late, not transported slowly
331 ms / 976 ms	mainnet median-node transit / p90 spread	median arrival – first sighting; p90 – min across sentries
0.027 → 0.051 ETH	block value at t=0 vs t≈3.5s (then flat)	MEV relay bid traces; V(t) plateaus when a block can't beat the deadline
7.04e-6 ETH/ms	dV/dt — value of one ms of delay	slope of V(t) over 1–4s
188 / 214,712 = 0.088%	orphan rate; hazard 1.6%→9.9%→28.5% across 4.0–6.0s	seen-blocks anti-join vs canonical chain
$35.10	modeled uplift per validator/yr at 6× (A 2.48 + C 32.76 + D 0.68; B 0.14 loses to C)	ETH @ $1,805.50, post-Pectra 46.2 ETH validators
~$4.8M / yr	all seven named partners at 6×	Kiln $1.55M, P2P $1.04M, Everstake $0.96M…
40% / 55%	adoption sweet spots (per-adopter / total value)	ramp `min(1, α/40%)` × decay `(1−α)`; θ=40% from proposer boost
16.8%	the seven partners' share of the network	147,737 / 880,550 validators — below the sweet spot
6.7× vs 2.2×	what a 150ms floor buys on Hoodi vs mainnet median	floor model: same product, different baselines
+11%	value gain going 6× → 20×	deadline headroom caps C; publication floor caps A
5.8%	proposals that capture max block value safely today	publish ≥95% of plateau AND arrive <4s
2.91% / +1.7%	their baseline APR / their own calculator's uplift	0.0159 ETH per 32 ETH — converges with our number
5 designs, 0 detections	the Hoodi identification record	ITS-vs-control, operator stepping, peer races, size-penalty DiD, mixture

Part 3 · The Statistics, MIDS-Style

3.1 Why the mainnet study is causal (the W241 story)

Two ingredients make this a natural experiment, not a correlation:

1. **A sharp exogenous threshold.** The spec — not any actor — sets the attestation deadline at 4,000ms. Blocks arriving at 3,950 vs 4,050ms are alike in every covariate; only votability changes. That's the regression discontinuity logic: identification comes from *local* comparison at a cutoff nobody can precisely manipulate (proposers can't steer gossip to the millisecond — and we ran a McCrary-style density check).
2. **Random assignment of the exposed.** RANDAO assigns attesters to slots by lottery. In potential-outcomes terms: treatment (being on duty when a late block appears) is independent of potential outcomes. So we compare *the same validators* across slots that randomly had early vs late blocks — not fast operators vs slow operators, which is selection on everything.

Say it like this: "*I never compare operators to each other — that's confounded to death. I compare the same validators across slots that, by lottery, got early or late blocks. RANDAO is my randomizer; the spec deadline is my discontinuity.*"

The honest limitation (volunteer it before they find it): our running variable is a *sentry-median* arrival, not each attester's own arrival. Classical measurement error in an RD's running variable smooths the jump — so sharp-RD point estimates are bandwidth-sensitive (τ ranged –0.005 to –0.124 across bandwidths). That's why we **lead with the dose-response** (the full measured curve), which needs no sharpness assumption, and treat the RD as supporting. Measurement error attenuates → our effects are lower bounds.

3.2 Why their regression is not causal

Their APR-uplift comes from a cross-section: operators with lower latency have higher APR. That's `selection on unobservables`: latency correlates with operator quality (relay connections, MEV tuning, timing strategy, geography). The latency coefficient absorbs all of it — classic omitted-variable bias. Then they multiply a Hoodi-measured Δ into that biased slope. Two errors, multiplied.

3.3 The DiD story (tell it as a journey)

1. **The dream:** staggered DiD of mump2p adoption on mainnet, Callaway–Sant'Anna estimator. C-S generalizes 2×2 DiD to staggered adoption: it estimates ATT(g,t) per adoption-cohort g and time t, using not-yet-treated units as controls, then aggregates — avoiding the negative-weighting bug in two-way fixed effects when effects vary over time.
2. **Why it died:** no staggering (all seven partners named in ONE press release, 2025-06-24) and no treatment (mump2p never reached mainnet — the treatment indicator is identically zero in the data). You cannot estimate the effect of a treatment that never occurred in your sample.
3. **The honest substitutes on Hoodi** (the only treated network), five designs:
   - **ITS + comparison series** (Hoodi vs never-treated Sepolia). One unit per arm — so *not* a real DiD; no cross-sectional variance, parallel trends untestable. The control still earns its keep: the biggest "effect" (–56% transit, 2025-09-29) appeared on the *control* 3 days later, bigger and cleaner (–72%, r^2 0.84 vs 0.34) → common shock, not treatment.
   - **Per-operator stepping**: segment proposers by fee-recipient (operators deposited in contiguous index batches), estimate a least-squares changepoint per cluster. Adoption = staggered breaks. Result: all 14 clusters break the *same day* — span 0 days → common shock.
   - **Peer-level races**: libp2p first-delivery wins per sending peer. Gateways would be new, fast, persistent winners. Result: young peers are no faster (–84 to +14ms edge); dominant winners born *after* the window.

- **Size-penalty DiD (label-free)**: second difference = block difficulty, not identity. A constant-time overlay must compress the big-block penalty. Result: **inverted** — penalty tripled (225→1,346ms) in the deployment window. That's a congestion signature (consistent with shadow-mode traffic duplication), the opposite of offloading.
- **Mixture emergence (label-free)**: if an unknown fraction adopted, log-transit becomes a 2-component mixture; the mixing weight *is* the adoption share. Fit by EM, gated on ΔBIC>10. Result: bimodality exists *before* deployment (the placebo fires) and no overlay-speed component ever appears.

Say it like this: "*When treatment assignment is unobservable, you move the second difference onto something you can observe — block difficulty, or the shape of the distribution. Five designs, including two that need no labels at all, and none finds the treatment. The absence isn't a labeling problem.*"

3.4 Inference: why not just OLS standard errors?

Slots within a day share network conditions (blob demand, client releases), so errors are correlated within day — 216k slots are far fewer *effective* observations. Fixes, in order:

- **Cluster-robust (CR1) sandwich**: allow arbitrary correlation within cluster, independence across; G=30 day-clusters.
- **Wild cluster bootstrap (WCR, Rademacher)**: with few clusters, CRVE over-rejects. Impose the null, flip cluster residual signs at random, rebuild the t-distribution empirically (Cameron-Gelbach-Miller). Caveat we cite: MacKinnon-Webb — even WCB misbehaves with few *treated* clusters.
- **Test discipline**: our suite includes a *size* test (under a true null, p-values ≈ uniform; nominal 10% test rejects ≈10%) and planted-effect recovery. An estimator you haven't watched fail on nulls isn't tested.

3.5 The counterfactual model — and the three bugs (tell these!)

Channels: **A** attester head votes (receive-side), **B** reorgs avoided, **C** MEV delay budget, **D** bandwidth. Total = A + max(B,C) + D.

The three bugs make you credible because you found them yourself:

1. **The tail-attester bug.** First model: a block only costs attesters if its *median* arrival is past 4s. Wrong — arrival is a *distribution* across nodes; 61% of real misses occur on on-time blocks (tail nodes). Fix: the measured dose-response f(arrival) *is* the tail-mass function; acceleration moves each slot along the measured curve. Lesson: never collapse a distribution to a point when the outcome lives in the tail.
2. **Baseline ≠ counterfactual estimator mismatch.** We briefly compared a *measured* network-average baseline to a *modeled* single-node counterfactual — different populations, invalid delta. Fix: both sides through the same estimator; speedup=1 must reproduce the measured rate (it does: 1.05% vs 0.94%). Lesson: a counterfactual that can't recover reality at the no-op is untrustworthy everywhere.
3. **The silent-zero Channel B.** Reorg pricing keyed on a column that didn't exist in the cached panel; a fallback returned $0.00 into a results table. Physically impossible (188 orphans measured!). Fix: measure the orphan hazard from *seen* blocks (winners AND losers — canonical-only data has survivorship bias), and add tests forbidding exact zeros. Lesson: silent fallbacks are how impossible numbers reach tables.

Also: **B and C are mutually exclusive** uses of the same saved milliseconds — bank them as earlier arrival (safety) or spend them as later publication (MEV), never both. Summing would double-count.

3.6 The transfer model (floor, not multiplier)

An RLNC overlay delivers in near-constant time; gossipsub scales with mesh depth. So the product is an **absolute floor**, and it runs in parallel: `transit_cf = min(observed, floor)`. The same 150ms floor = 6.7× on Hoodi (baseline ~1,000ms) but 2.2× at mainnet's median node (331ms) — and 0.8×, a *downgrade*, if the realized floor is 400ms. Multipliers don't transfer across baselines; floors do. Bonus: 20× multiplicative on mainnet implies 17ms global propagation — faster than light in fiber.

3.7 The adoption sweep (the game theory)

Channel C per adopter = C_max × min(1, α/θ) × (1−α):

- **Ramp**: a proposer can only *spend* delay if ~θ=40% of attesters receive fast (proposer-boost reorg threshold). Below that, delaying is suicide.
- **Decay**: the captured MEV is flow stolen from the *next* slot; with probability α your predecessor steals from you the same way. Red-queen race.
- Peaks: per-adopter at α=θ (40%), total value at ~55%, → public-goods floor (3.30)*at*10032 and never decays — late adoption is bought to stop bleeding, not to gain.

Part 4 · Spoken Scripts (memorize verbatim)

The 30-second version

"Optimum markets 6-to-20× faster block propagation. I measured what that's actually worth on Ethereum mainnet using public data: about thirty-five dollars per validator per year — and their own APR calculator, buried in a Notion doc, agrees with me within twenty percent. Ninety-two percent of that value is MEV timing, not reliability, and it only pays if you re-tune your block publication. On Hoodi — the only network where the product runs — I tried five independent ways to detect it, including two that need no adopter labels, and found nothing. The value is real but modest, front-loaded to partial adoption, and their headline number was benchmarked against the slowest gossipsub conditions on their slowest network."

The 2-minute version (add these beats)

1. **The physics**: latency is free until 4,000ms, then a cliff — 99.9% to 67% correct-head votes. Mean latency is the wrong KPI; the tail is everything. 61% of attestation misses happen on blocks that arrived *on time* — they're nodes in the propagation tail.
2. **The reframe**: late blocks are *published* late (3.65s vs 1.8s), not transported slowly. Even an infinitely fast network rescues only ~80% of them. So Optimum isn't selling throughput — it's selling *delay budget*: publish later at the same arrival, catch a richer MEV bid. dV/dt ≈ 7 micro-ETH per millisecond.
3. **The economics**: $35/validator/yr at their 6×; going to their 20× adds only 11% because the deadline caps the budget. Sweet spot at 40–55% adoption; their seven partners are at 16.8% — below it. At saturation the MEV channel cannibalizes to zero and only public goods remain.
4. **The identification**: mainnet effects identified off the spec deadline plus RANDAO's random committee assignment — never operator-vs-operator.

Hoodi tested with five designs; every clean break belonged to the never-treated control; the one non-null (size-penalty DiD) was *inverted* — congestion during their benchmark window, not offloading.

5. **The close**: "*their calculator and my study agree on the magnitude; the difference is my number survives cross-examination.*"

Answers to the questions you WILL get

"Isn't your RD invalid? Blob count jumps at the cutoff." "*Correct — which is why I don't lead with the sharp RD. The running variable is a sentry median, so measurement error smooths the discontinuity and makes τ bandwidth-sensitive; covariate balance fails on blob count. I report that openly and rest the claim on the dose-response, which needs no sharpness assumption, and on RANDAO for exogeneity. Attenuation also means my estimates are conservative.*"

"Why is the attester channel so small? Faster propagation helps attestations!" "*It helps them by pulling your node out of the propagation tail — worth about $2.50 a year per validator, because source and target rewards, 74% of the attestation slice, are structurally immune to latency. Only timely-head is exposed, the miss rate is under 1%, and most of what remains is late publication, which no transport fixes.*"

"Your Hoodi nulls just mean shadow mode. The product still works." "*Agreed — I say exactly that: absence of evidence under shadow mode, not proof of absence. But it cuts both ways: there is then no deployment evidence anywhere, and every quantitative claim rests on gateway self-telemetry from a degraded testnet window. I priced the claim generously assuming it holds; that's the $35.*"

"Why should anyone believe your counterfactual over their measurement?" "*Because their measurement is of a confounded cross-section and mine is of a randomized natural experiment — and because when I force my model to predict reality with the treatment turned off, it reproduces the measured baseline. Their slope can't run that check. And note we agree on the magnitude anyway.*"

"Mean-field is standard. Why does adoption dynamics matter?" "*Because the value is a relative advantage in a timing race. Below ~40% adoption you can't safely spend the delay; above it, adopters steal from each other one-for-one. Mean-field prices the one world that can't exist — a lone adopter with full network support. The observable consequence: their partners, at 16.8% of stake, are below the sweet spot — which is actually a sales argument they're missing.*"

Part 5 · Flash Cards

Q	A
Attestation deadline?	4,000ms — 1/3 of the slot, spec-set
Correct-head at 4–5s?	67.4% (vs 99.9% on time)
Share of misses on on-time blocks?	61% — tail nodes
Late blocks' median publication?	3,656ms (normal: 1,811ms)
dV/dt?	~7.0e-6 ETH/ms; V plateaus 0.051 ETH at ~3.5s
Orphan rate / hazard cliff?	0.088%; 1.6% → 9.9% → 28.5% over 4.0–6.0s
Per-validator value at 6×?	$35.10 (C=32.76, A=2.48, D=0.68, B loses to C)
6× → 20× gains?	+11% — deadline headroom binds
Floor model?	transit = min(observed, floor); 150ms = 6.7× Hoodi, 2.2× mainnet median
Adoption sweet spots?	40% per-adopter (20.82), 557.95M); partners at 16.8%
Hoodi designs & result?	5 designs, 0 detections; size-penalty DiD inverted (congestion)
Their calculator's uplift?	0.0159 ETH per 32 ETH ≈ +1.7% APR ≈ our number ±20%
Why not compare operators?	Selection: latency proxies operator quality → OVB
Why WCR bootstrap?	Few clusters → CRVE over-rejects; impose null, Rademacher flips
B/C rule?	Mutually exclusive uses of the same ms: total = A + max(B,C) + D
The three bugs?	Tail-attester (distribution ≠ point); estimator mismatch (1× must recover baseline); silent-zero B (survivorship + fallback)

Sources: 216k-slot mainnet panel (Xatu, 2025-06), 487 network-days Hoodi/Sepolia, MEV relay bid traces, 1.6M-proposal operator scan, libp2p peer races. Pipeline: this repo; live: brian.biz/optimum. Every dollar figure is a modeled counterfactual — mump2p is not deployed on mainnet.